Math 54 — Week 8

Diagonalization; eigenvectors and linear transformations; complex eigenvalues (Lay 5.3–5.5)

§1 Facts used from Weeks 1–7

(W1) **Matrix times matrix, column by column.** $AP = [\,A\mathbf{p}_1 \;\cdots\; A\mathbf{p}_n\,]$, where $\mathbf{p}_j$ are the columns of P. And P times a diagonal matrix $D = \operatorname{diag}(d_1, \ldots, d_n)$ multiplies column j of P by d_j: $PD = [\,d_1\mathbf{p}_1 \;\cdots\; d_n\mathbf{p}_n\,]$.

(W2) **Invertible matrix theorem.** For an $n \times n$ matrix P: P is invertible $\iff$ its columns are linearly independent $\iff$ its columns form a basis of $\mathbb{R}^n$ $\iff$ $\det P \neq 0$.

(W3) **Eigenvalues** (Week 7). λ is an eigenvalue of A $\iff$ $\det(A - \lambda I) = 0$; the eigenspace for λ is $\operatorname{Nul}(A - \lambda I)$. Eigenvectors for distinct eigenvalues are linearly independent. Similar matrices ($B = P^{-1}AP$) have the same characteristic polynomial.

(W4) **Coordinates** (Week 7). If $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ is a basis of $\mathbb{R}^n$ and $P_{\mathcal{B}} = [\,\mathbf{b}_1 \cdots \mathbf{b}_n\,]$, then $\mathbf{x} = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$ and $[\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}$. Taking coordinates is linear and one-to-one.

(W5) **Cofactor expansion** (Week 5). A determinant can be expanded down any column; if a column has a single nonzero entry c in position (i, j), the determinant is $(-1)^{i+j}c$ times the minor.

§2 This week

§2.1 Diagonalization (Lay 5.3)

The question. The difference equation $\mathbf{x}_{k+1} = A\mathbf{x}_k$ has the solution $\mathbf{x}_k = A^k\mathbf{x}_0$. Computing A^{100} by multiplying is hopeless. When can A^k be written down directly for every k?

Powers of a diagonal matrix are easy: $D = \operatorname{diag}(d_1, \ldots, d_n)$ has $D^k = \operatorname{diag}(d_1^k, \ldots, d_n^k)$, because multiplying diagonal matrices multiplies their diagonal entries. The same is true for any matrix of the form $A = PDP^{-1}$: in the product $A^k = (PDP^{-1})(PDP^{-1})\cdots(PDP^{-1})$ each inner $P^{-1}P$ cancels, so

$$A^k = PD^kP^{-1}.$$

Ex. 1. Let $A = \left[\begin{smallmatrix} 7 & 2 \\ -4 & 1 \end{smallmatrix}\right]$. With

$$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix},$$

one checks $A = PDP^{-1}$. Therefore

$$A^k = P\begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix}P^{-1} = \begin{bmatrix} 2\cdot 5^k - 3^k & 5^k - 3^k \\ 2\cdot 3^k - 2\cdot 5^k & 2\cdot 3^k - 5^k \end{bmatrix}.$$

(For $k = 1$ this gives back A.) Where did P and D come from? The columns of P are eigenvectors of A: $A(1,-1) = (5,-5)$ and $A(1,-2) = (3,-6)$. The theorem below says this is the only way such a factorization can arise.

Definition. A square matrix A is **diagonalizable** if $A = PDP^{-1}$ for some invertible P and diagonal D; that is, if A is similar to a diagonal matrix.

Theorem (Diagonalization theorem (Lay Thm. 5))

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. In that case $A = PDP^{-1}$ exactly when the columns of P are n independent eigenvectors and the diagonal entries of D are the corresponding eigenvalues, in the same order.

The idea

Rewrite $A = PDP^{-1}$ as $AP = PD$ and compare the two sides column by column (W1). Column j of AP is $A\mathbf{p}_j$; column j of PD is $d_j\mathbf{p}_j$. So $AP = PD$ says exactly that each column of P is an eigenvector with eigenvalue d_j. The only extra requirement, that P be invertible, says the columns are independent.

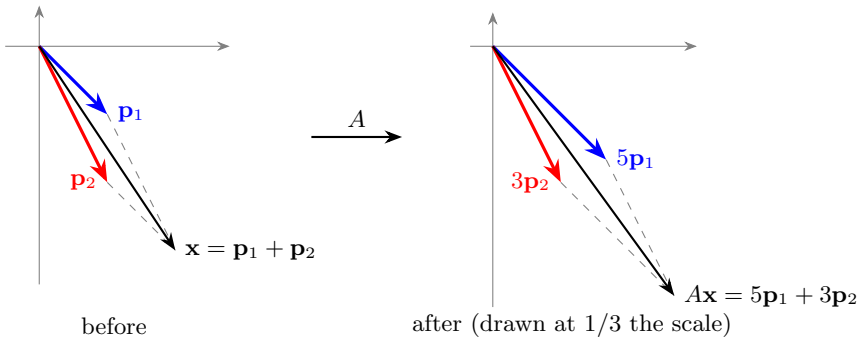
Proof. 1. Let $P = [\,\mathbf{p}_1 \cdots \mathbf{p}_n\,]$ be any $n \times n$ matrix and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. By (W1),

$$AP = [\,A\mathbf{p}_1 \ \cdots \ A\mathbf{p}_n\,], \qquad PD = [\,\lambda_1\mathbf{p}_1 \ \cdots \ \lambda_n\mathbf{p}_n\,].$$

2. ($\Rightarrow$) Suppose $A = PDP^{-1}$. Multiplying on the right by P gives $AP = PD$, so $A\mathbf{p}_j = \lambda_j\mathbf{p}_j$ for each j. P is invertible, so its columns are independent (W2); in particular none is $\mathbf{0}$. So the $\mathbf{p}_j$ are n independent eigenvectors, with eigenvalues λ_j.

3. ($\Leftarrow$) Suppose $\mathbf{p}_1, \ldots, \mathbf{p}_n$ are independent eigenvectors, $A\mathbf{p}_j = \lambda_j\mathbf{p}_j$. By step 1, $AP = PD$. The columns of P are independent, so P is invertible (W2). Multiplying $AP = PD$ on the right by P^{-1} gives $A = PDP^{-1}$.

□



For A of Ex. 1: A stretches $\mathbf{p}_1$ by 5 and $\mathbf{p}_2$ by 3. Any $\mathbf{x}$ is a combination of $\mathbf{p}_1$ and $\mathbf{p}_2$, and A acts on each piece separately, so $A^k\mathbf{x} = 5^k\mathbf{p}_1 + 3^k\mathbf{p}_2$ when $\mathbf{x} = \mathbf{p}_1 + \mathbf{p}_2$.

Diagonalizing a matrix

1. Find the eigenvalues: solve $\det(A - \lambda I) = 0$.
2. For each eigenvalue, find a basis of the eigenspace $\operatorname{Nul}(A - \lambda I)$.
3. If the bases together have n vectors, put them in P as columns; otherwise A is not diagonalizable (theorems below).
4. Put the eigenvalues in D in the same order as the columns of P. Check $AP = PD$ (this avoids computing P^{-1}).

Ex. 2: a repeated eigenvalue that still works. Let

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}, \qquad \det(A - \lambda I) = -(\lambda - 1)(\lambda + 2)^2.$$

1. Eigenvalues 1 and -2 (algebraic multiplicity 2).
2. $A - I$ row reduces to two pivots, and $\mathrm{Nul}(A - I)$ has basis $\mathbf{v}_1 = (1, -1, 1)$. $A + 2I$ has all three rows multiples of $(1, 1, 1)$, so $\mathrm{Nul}(A + 2I)$ is the plane $x_1 + x_2 + x_3 = 0$, with basis $\mathbf{v}_2 = (-1, 1, 0)$, $\mathbf{v}_3 = (-1, 0, 1)$.
3. Three vectors in all, so
$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix},$$
and $AP = PD$. ($\det P = 1 \neq 0$ confirms that the columns are independent.)

Ex. 3: the same eigenvalues, not diagonalizable. For
$$A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix},$$
again $\det(A - \lambda I) = -(\lambda - 1)(\lambda + 2)^2$, but $A + 2I$ has rank 2, so the eigenspace for -2 is only a line. Each eigenspace is at most what it is, so there are at most $1 + 1 = 2$ independent eigenvectors, and A is not diagonalizable. Knowing the eigenvalues is not enough; the eigenspace dimensions decide.

Theorem (n distinct eigenvalues are enough (Lay Thm. 6))
An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Proof. Pick one eigenvector for each of the n eigenvalues. Eigenvectors for distinct eigenvalues are independent (W3), so these are n independent eigenvectors, and the diagonalization theorem applies. □

The condition is sufficient, not necessary: the matrix of Ex. 2 has only two distinct eigenvalues and is diagonalizable, and I has one.

Theorem (An eigenspace is no bigger than the multiplicity (Lay Thm. 7a))
If λ_0 is an eigenvalue of A with algebraic multiplicity m, then $1 \leq \dim \mathrm{Nul}(A - \lambda_0 I) \leq m$.

The idea

Use the eigenvectors as the first columns of a new basis. In that basis the matrix of A starts with λ_0 on the diagonal and zeros below it in those columns, and a determinant of such a matrix visibly has the factor $(\lambda_0 - \lambda)$ once for each of those columns. Similar matrices have the same characteristic polynomial, so that factor is in A's polynomial too.

Proof. 1. The eigenspace contains an eigenvector, so its dimension p is at least 1. Let $\mathbf{v}_1, \ldots, \mathbf{v}_p$ be a basis of it, and extend to a basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$ of $\mathbb{R}^n$ (or $\mathbb{C}^n$). Let $P = [\,\mathbf{v}_1 \cdots \mathbf{v}_n\,]$, invertible by (W2), and $B = P^{-1}AP$.

2. For $j \leq p$, $P^{-1}\mathbf{v}_j$ is the coordinate vector of $\mathbf{v}_j$ in the new basis, which is $\mathbf{e}_j$. So column j of B is
$$P^{-1}A\mathbf{v}_j = \lambda_0 P^{-1}\mathbf{v}_j = \lambda_0 \mathbf{e}_j.$$
So the first p columns of $B - \lambda I$ have a single nonzero entry, $\lambda_0 - \lambda$, on the diagonal.

3. Expand $\det(B - \lambda I)$ down column 1, then the resulting minor down its column 1, and so on, p times (W5). Each step contributes a factor $\lambda_0 - \lambda$, so $\det(B - \lambda I) = (\lambda_0 - \lambda)^p\, q(\lambda)$ for a polynomial q.

4. B is similar to A, so $\det(A - \lambda I) = \det(B - \lambda I)$ (W3), and $(\lambda - \lambda_0)$ appears in it at least p times: $m \geq p$.

□

Theorem (When the eigenvectors are enough (Lay Thm. 7b,c))

Let A be $n \times n$ with distinct eigenvalues $\lambda_1, \dots, \lambda_r$.

(a) If $\mathcal{B}_k$ is a basis of the eigenspace for λ_k, the vectors of $\mathcal{B}_1, \dots, \mathcal{B}_r$ together are linearly independent.

(b) A is diagonalizable if and only if the dimensions of the eigenspaces add up to n, if and only if the characteristic polynomial factors completely into linear factors and the dimension of each eigenspace equals the algebraic multiplicity of its eigenvalue.

The idea

(a) is the distinct-eigenvalue independence theorem applied to groups instead of single vectors: in a relation, add up the terms from each eigenspace. Each group sum is an eigenvector or $\mathbf{0}$, and eigenvectors for distinct eigenvalues cannot cancel. (b) is counting: there are n independent eigenvectors exactly when the eigenspaces supply n vectors, and by the previous theorem each eigenspace supplies at most its multiplicity, while the multiplicities add up to at most n.

Proof. 1. (a) Take a relation among all the vectors, and let $\mathbf{w}_k$ be the sum of the terms coming from $\mathcal{B}_k$. Then $\mathbf{w}_1 + \cdots + \mathbf{w}_r = \mathbf{0}$, and each $\mathbf{w}_k$ is in the eigenspace for λ_k.

2. If some $\mathbf{w}_k$ were nonzero, the nonzero ones would be eigenvectors for distinct eigenvalues with sum $\mathbf{0}$, a dependence among them, which (W3) rules out. So every $\mathbf{w}_k = \mathbf{0}$.

3. $\mathbf{w}_k = \mathbf{0}$ is a relation among the vectors of the basis $\mathcal{B}_k$, so all its coefficients are 0. This holds for every k, so the original relation was trivial.

4. (b) Let d_k be the eigenspace dimensions and m_k the multiplicities. Every eigenvector lies in one eigenspace, so by (a) the most independent eigenvectors one can have is $d_1 + \cdots + d_r$. By the diagonalization theorem, A is diagonalizable $\iff d_1 + \cdots + d_r = n$.

5. The roots λ_k of the characteristic polynomial account for $m_1 + \cdots + m_r \le n$ of its n linear factors, with equality exactly when it factors completely. Since $d_k \le m_k$, the sum $d_1 + \cdots + d_r$ equals n exactly when $m_1 + \cdots + m_r = n$ and each $d_k = m_k$.

□

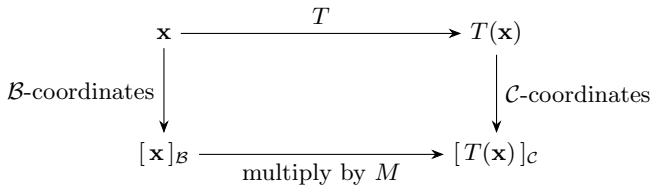
§2.2 Eigenvectors and linear transformations (Lay 5.4)

The question. What does $A = PDP^{-1}$ say about the transformation $\mathbf{x} \mapsto A\mathbf{x}$ itself? And how do we describe a linear transformation on a space like $\mathbb{P}_2$ by a matrix?

The matrix of a linear transformation. Let $T\colon V \to W$ be linear, $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ a basis of V and $\mathcal{C}$ a basis of W. Write $\mathbf{x} = r_1\mathbf{b}_1 + \cdots + r_n\mathbf{b}_n$. By linearity $T(\mathbf{x}) = r_1T(\mathbf{b}_1) + \cdots + r_nT(\mathbf{b}_n)$, and taking $\mathcal{C}$-coordinates is linear (W4), so

$$[T(\mathbf{x})]_{\mathcal{C}} = r_1[T(\mathbf{b}_1)]_{\mathcal{C}} + \cdots + r_n[T(\mathbf{b}_n)]_{\mathcal{C}} = M[\mathbf{x}]_{\mathcal{B}}, \qquad M = \big[\,[T(\mathbf{b}_1)]_{\mathcal{C}} \ \cdots \ [T(\mathbf{b}_n)]_{\mathcal{C}}\,\big].$$

M is the **matrix for T relative to $\mathcal{B}$ and $\mathcal{C}$**: its columns are the images of the $\mathcal{B}$ vectors, written in $\mathcal{C}$-coordinates. When $W = V$ and $\mathcal{C} = \mathcal{B}$, M is the **$\mathcal{B}$-matrix** of T, written $[T]_{\mathcal{B}}$, and $[T(\mathbf{x})]_{\mathcal{B}} = [T]_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$.



Going across the top (T, then coordinates) gives the same result as going down and across the bottom (coordinates, then M). M is the transformation, translated into coordinates.

Ex.: differentiation on $\mathbb{P}_2$. Let $T(a_0 + a_1 t + a_2 t^2) = a_1 + 2a_2 t$ (the derivative), and $\mathcal{B} = \{1, t, t^2\}$. Then $T(1) = 0$, $T(t) = 1$, $T(t^2) = 2t$, with coordinate vectors $(0,0,0)$, $(1,0,0)$, $(0,2,0)$:

$$[T]_{\mathcal{B}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \qquad [T]_{\mathcal{B}} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ 2a_2 \\ 0 \end{bmatrix},$$

which is the coordinate vector of $a_1 + 2a_2 t$, as it should be.

Theorem (Diagonal matrix representation (Lay Thm. 8))

Suppose $A = PCP^{-1}$, with P invertible, and let $\mathcal{B}$ be the basis of $\mathbb{R}^n$ formed by the columns of P. Then C is the $\mathcal{B}$-matrix of the transformation $\mathbf{x} \mapsto A\mathbf{x}$. In particular, if $A = PDP^{-1}$ with D diagonal, then D is the $\mathcal{B}$-matrix of $\mathbf{x} \mapsto A\mathbf{x}$.

The idea

In $\mathcal{B}$-coordinates, $\mathbf{x}$ is $P^{-1}\mathbf{x}$ (W4). So the transformation in $\mathcal{B}$-coordinates is: convert back with P, apply A, convert again with P^{-1}; that is, multiplication by $P^{-1}AP = C$. For a diagonal D this says: in the eigenvector basis, the transformation just scales each coordinate.

Proof. 1. By (W4), $\mathbf{x} = P[\mathbf{x}]_{\mathcal{B}}$ and $[A\mathbf{x}]_{\mathcal{B}} = P^{-1}(A\mathbf{x})$.

2. So $[A\mathbf{x}]_{\mathcal{B}} = P^{-1}AP[\mathbf{x}]_{\mathcal{B}}$. Since $A = PCP^{-1}$, $P^{-1}AP = C$.

3. The matrix that converts $[\mathbf{x}]_{\mathcal{B}}$ into $[A\mathbf{x}]_{\mathcal{B}}$ is the $\mathcal{B}$-matrix, so $[\mathbf{x} \mapsto A\mathbf{x}]_{\mathcal{B}} = C$.

□

- For A of Ex. 1 and $\mathcal{B} = \{(1,-1),(1,-2)\}$: the $\mathcal{B}$-matrix is $\mathrm{diag}(5,3)$. In $\mathcal{B}$-coordinates, A multiplies the first coordinate by 5 and the second by 3.
- **A non-diagonalizable example.** $A = \left[\begin{smallmatrix} 4 & -9 \\ 4 & -8 \end{smallmatrix}\right]$ has the single eigenvalue -2 with a one-dimensional eigenspace, spanned by $(3,2)$, so it is not diagonalizable. With $\mathcal{B} = \{(3,2),(2,1)\}$:

  $$[\mathbf{x} \mapsto A\mathbf{x}]_{\mathcal{B}} = P^{-1}AP = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}.$$

  Not diagonal, but triangular: the first basis vector is an eigenvector, and the second is sent to -2 times itself plus the first.
- Similar matrices are matrices of the same transformation in different bases. This is why they share eigenvalues and the characteristic polynomial: those are properties of the transformation, not of the basis.

§2.3 Complex eigenvalues (Lay 5.5)

The question. Rotation by 90° moves every nonzero vector off its line, so it has no real eigenvectors, and $\det(A - \lambda I) = \lambda^2 + 1$ has no real roots. What do the complex roots $\pm i$ mean geometrically?

Complex eigenvalues. Allow complex scalars: $\lambda \in \mathbb{C}$ is a (complex) eigenvalue of A if $A\mathbf{x} = \lambda\mathbf{x}$ for some nonzero $\mathbf{x} \in \mathbb{C}^n$. Everything in Week 7 works unchanged over $\mathbb{C}$; in particular the eigenvalues are the complex roots of $\det(A - \lambda I)$, and every $n \times n$ matrix has n of them, counted with multiplicity. For a complex vector $\mathbf{x}$, $\bar{\mathbf{x}}$ conjugates every entry, and $\mathrm{Re}\,\mathbf{x}$, $\mathrm{Im}\,\mathbf{x}$ are the real vectors with $\mathbf{x} = \mathrm{Re}\,\mathbf{x} + i\,\mathrm{Im}\,\mathbf{x}$.

Ex.: rotation by 90°. $A = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$, $\det(A - \lambda I) = \lambda^2 + 1$, eigenvalues $\pm i$. For $\lambda = i$, the first row of $A - iI$ gives $-ix_1 - x_2 = 0$, so $x_2 = -ix_1$ and $\mathbf{x} = (1, -i)$. Check: $A(1,-i) = (i,1)$, and $i(1,-i) = (i,1)$.

- **Conjugate pairs.** If A is real and $A\mathbf{x} = \lambda\mathbf{x}$, then conjugating both sides gives $A\bar{\mathbf{x}} = \bar{\lambda}\bar{\mathbf{x}}$ (the entries of A are real, so they are unchanged). So complex eigenvalues of a real matrix come in pairs $\lambda, \bar{\lambda}$, with conjugate eigenvectors; there is no need to compute the second one.

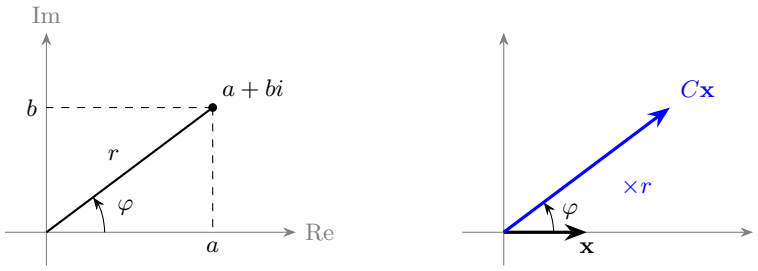
Rotation–scaling matrices. For real a, b, not both 0, let

$$C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}, \qquad \det(C - \lambda I) = (a - \lambda)^2 + b^2, \qquad \lambda = a \pm bi.$$

Write $r = |\lambda| = \sqrt{a^2 + b^2}$ and let φ be the angle of the point (a, b), so $a = r\cos\varphi$, $b = r\sin\varphi$. Then

$$C = r\begin{bmatrix}\cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi\end{bmatrix}:$$

C rotates by φ and scales by r. The eigenvalues $a \pm bi$ carry both numbers: the modulus is the scale factor and the argument is the angle.



Left: the eigenvalue $a + bi$ in the complex plane, with modulus r and angle φ. Right: C turns every vector by φ and multiplies its length by r; here $C\mathbf{e}_1 = (a, b)$.

Theorem (Every real 2×2 matrix with complex eigenvalues is a rotation–scaling in disguise (Lay Thm. 9))
Let A be a real 2×2 matrix with a complex eigenvalue $\lambda = a - bi$ $(b \neq 0)$ and an eigenvector $\mathbf{v}$ in $\mathbb{C}^2$. Then $A = PCP^{-1}$, where

$$P = [\operatorname{Re}\mathbf{v} \quad \operatorname{Im}\mathbf{v}], \qquad C = \begin{bmatrix}a & -b \\ b & a\end{bmatrix}.$$

The idea

Write $\mathbf{v} = \mathbf{x} + i\mathbf{y}$ with real $\mathbf{x}, \mathbf{y}$. The single complex equation $A\mathbf{v} = \lambda\mathbf{v}$ is two real equations: its real part and its imaginary part. Each says what A does to $\mathbf{x}$ or to $\mathbf{y}$, as a combination of $\mathbf{x}$ and $\mathbf{y}$, and the coefficients turn out to be the entries of C. Then $AP = PC$, as in the diagonalization theorem but with C in place of D.

Proof. 1. Let $\mathbf{x} = \operatorname{Re}\mathbf{v}$ and $\mathbf{y} = \operatorname{Im}\mathbf{v}$, so $\mathbf{v} = \mathbf{x} + i\mathbf{y}$. Since A is real, $A\mathbf{v} = A\mathbf{x} + iA\mathbf{y}$ with $A\mathbf{x}$, $A\mathbf{y}$ real.

2. Expand the right side:
$$\lambda\mathbf{v} = (a - bi)(\mathbf{x} + i\mathbf{y}) = (a\mathbf{x} + b\mathbf{y}) + i(-b\mathbf{x} + a\mathbf{y}).$$

3. Real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ must agree, so $A\mathbf{x} = a\mathbf{x} + b\mathbf{y}$ and $A\mathbf{y} = -b\mathbf{x} + a\mathbf{y}$.

4. By (W1), these two equations say
$$AP = [\,A\mathbf{x} \;\; A\mathbf{y}\,] = [\,\mathbf{x} \;\; \mathbf{y}\,]\begin{bmatrix}a & -b \\ b & a\end{bmatrix} = PC.$$

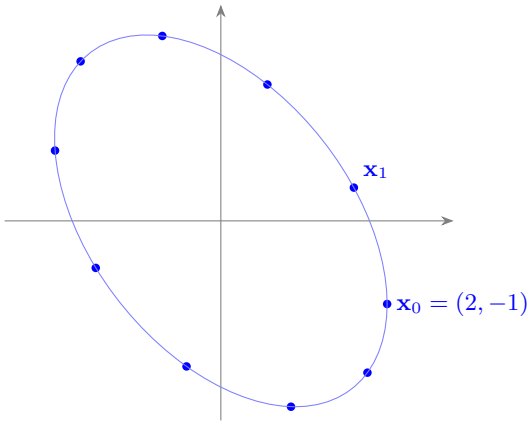
5. *P is invertible.* If $\mathbf{x}, \mathbf{y}$ were dependent, $\mathbf{v}$ would be a complex multiple of a single real vector $\mathbf{u} \neq \mathbf{0}$ (one of $\mathbf{x}, \mathbf{y}$). That $\mathbf{u}$ is then also an eigenvector for λ, so $A\mathbf{u} = \lambda\mathbf{u}$. The left side is real; the right side is not, because $\mathbf{u}$ is real and nonzero and λ is not real $(b \neq 0)$. So $\mathbf{x}, \mathbf{y}$ are independent, and $A = PCP^{-1}$.

□

Ex. 2 of Lay 5.5. Let $A = \left[\begin{smallmatrix}0.5 & -0.6 \\ 0.75 & 1.1\end{smallmatrix}\right]$.

1. $\det(A - \lambda I) = \lambda^2 - 1.6\lambda + 1$, so $\lambda = 0.8 \pm 0.6i$, and $|\lambda| = \sqrt{0.64 + 0.36} = 1$.
2. For $\lambda = 0.8 - 0.6i$, the first row of $A - \lambda I$ gives $(-0.3 + 0.6i)x_1 - 0.6x_2 = 0$, so $x_2 = (-0.5 + i)x_1$. With $x_1 = 2$: $\mathbf{v} = (2, -1 + 2i)$.
3. $\operatorname{Re}\mathbf{v} = (2, -1)$, $\operatorname{Im}\mathbf{v} = (0, 2)$, and $a = 0.8$, $b = 0.6$:
$$P = \begin{bmatrix}2 & 0 \\ -1 & 2\end{bmatrix}, \qquad C = \begin{bmatrix}0.8 & -0.6 \\ 0.6 & 0.8\end{bmatrix}, \qquad A = PCP^{-1}.$$

4. Here $r = 1$: C is a pure rotation (by the angle with $\cos\varphi = 0.8$, about 36.9°). So in the $\mathcal{B}$-coordinates given by the columns of P, A rotates; in standard coordinates the iterates $A^k\mathbf{x}_0$ travel around an ellipse.



The iterates $\mathbf{x}_k = A^k\mathbf{x}_0$ for $\mathbf{x}_0 = \operatorname{Re}\mathbf{v} = (2, -1)$. In the coordinates given by the columns of P each step is a rotation by about 36.9°; in standard coordinates the points move around an ellipse. If $|\lambda|$ were greater than 1 they would spiral outward, and if less than 1, inward.

Why $a - bi$ and not $a + bi$. Using the eigenvector for $\bar{\lambda} = a + bi$ instead gives $\bar{\mathbf{v}}$, whose imaginary part is $-\operatorname{Im}\mathbf{v}$; that changes P and the sign of b in C, so the rotation is described the other way around. Either works if used consistently; Lay's convention pairs $\lambda = a - bi$ with $C = \left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$.

Shortcuts

- **Powers**: $A = PDP^{-1}$ gives $A^k = PD^kP^{-1}$; for a limit as $k \to \infty$, only eigenvalues with $|\lambda| \geq 1$ survive.
- **Check a diagonalization** with $AP = PD$ (column j: $A\mathbf{p}_j = d_j\mathbf{p}_j$), never by computing P^{-1}.
- **Quick test**: n distinct eigenvalues ⇒ diagonalizable. A repeated eigenvalue λ of multiplicity m needs $\operatorname{rank}(A - \lambda I) = n - m$.
- **Triangular with repeated diagonal entries**: compute the rank of $A - \lambda I$ for the repeated λ; that decides it.
- **$\mathcal{B}$-matrix**: columns are $[T(\mathbf{b}_j)]_{\mathcal{B}}$. For $T(\mathbf{x}) = A\mathbf{x}$ on $\mathbb{R}^n$ it is $P^{-1}AP$ with $P = [\,\mathbf{b}_1 \cdots \mathbf{b}_n\,]$.
- **Complex eigenvector of a 2×2**: one row of $A - \lambda I$, say $(p \;\; q)$ with $q \neq 0$, gives the eigenvector $(q, -p)$ directly, or $(-q, p)$.
- **Rotation–scaling data**: from $\lambda = a \pm bi$, scale factor $r = \sqrt{a^2 + b^2} = \sqrt{\det A}$, angle from $\cos\varphi = a/r$.

Traps

- The columns of P and the diagonal of D must be in the same order.
- Diagonalizable is not the same as invertible: $\operatorname{diag}(1, 0)$ is diagonalizable and singular; $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ is invertible and not diagonalizable.
- A repeated eigenvalue does not by itself mean "not diagonalizable"; check the eigenspace dimension.
- Row reducing A changes its eigenvalues; row reduce $A - \lambda I$ only to find eigenvectors.
- In Thm. 9, P uses $\operatorname{Re}\mathbf{v}$ and $\operatorname{Im}\mathbf{v}$ (real vectors), not $\mathbf{v}$ and $\bar{\mathbf{v}}$, and the eigenvalue used is $a - bi$.
- Complex eigenvalues of a *real* matrix come in conjugate pairs; a complex matrix need not have that property.

§3 Homework this week

HW7, Section 5.3: 1, 3, 7, 13, 19, 22, 23, 24, 26, 30. Section 5.4: 1, 3, 4, 5, 7, 8, 9, 15, 20, 22, 23, 27. HW8, Section 5.5: 3, 5, 10, 13, 15, 24.

§4 Problem set

P1. Diagonalize $A = \left[\begin{smallmatrix} 2 & 3 \\ 4 & 1 \end{smallmatrix}\right]$, and find a formula for A^k.

P2. Decide whether each matrix is diagonalizable, and diagonalize it if it is:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 4 & 0 & -2 \\ 2 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix}.$$

P3. A 4×4 matrix has eigenvalues 1, 1, 2, 3, and $\operatorname{rank}(A - I) = 3$. Is A diagonalizable? What if $\operatorname{rank}(A - I) = 2$?

P4. Let $T\colon \mathbb{P}_1 \to \mathbb{P}_2$ be $T(p) = (t+1)p(t)$. Find the matrix of T relative to $\mathcal{B} = \{1, t\}$ and $\mathcal{C} = \{1, t, t^2\}$, and use it to compute $T(3 - 2t)$.

P5. Let $A = \left[\begin{smallmatrix} 3 & -1 \\ 2 & 0 \end{smallmatrix}\right]$ and $\mathcal{B} = \{(1,1), (1,2)\}$. Find the $\mathcal{B}$-matrix of $\mathbf{x} \mapsto A\mathbf{x}$. What does it tell you about $(1,1)$ and $(1,2)$?

P6. Let $A = \left[\begin{smallmatrix} 1 & -2 \\ 1 & 3 \end{smallmatrix}\right]$. Find its complex eigenvalues, an eigenvector for $2 - i$, and P, C with $A = PCP^{-1}$. By what factor does A scale, in the sense of C?

P7. Prove: if A is diagonalizable and invertible, then A^{-1} is diagonalizable. If A is diagonalizable, is A^{T}?

P8. Let $A = \left[\begin{smallmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{smallmatrix}\right]$ and $\mathbf{x}_0 = (1, 0)$. Find $\mathbf{x}_k = A^k \mathbf{x}_0$ and its limit as $k \to \infty$.